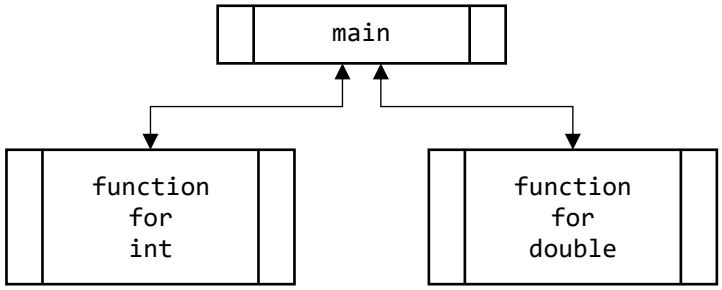
ECE 2305 – Introduction to C Programming
Programming Tutorial 19
Overloaded Functions

Write a C++ program to demonstrate user-defined functions and function calls as illustrated below. Both user-defined functions must have the same function name. This is an exercise in overloading the function name.



The functions are described below in "pseudo code."

function (return type void, argument type int, variable name *n*)
display " welcome to function for int"
display Memory Map for function
variable *n* address and value

function (return type void, argument type double, variable name *n*)
display " welcome to function for double"
display Memory Map for function
variable *n* address and value

main function:
function call with integer value
display "welcome back to main"
function call with double value
display "welcome back to main"

Document the exercise in a Word document.

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Input

```
// ECE 2305 - Tutorial 19

#include <iostream>
using namespace std;

void function(int n) {
    cout << "Welcome to function for int" << endl;
    cout << "Memory Address of n: " << &n << " Value: " << n << endl;
}

void function(double n) {
    cout << "Welcome to function for double" << endl;
    cout << "Memory Address of n: " << &n << " Value: " << n << endl;
}

int main() {
    int intValue = 10;
    double doubleValue = 5.5;

    function(intValue);
    cout << "Welcome back to main" << endl;

    function(doubleValue);
    cout << "Welcome back to main" << endl;

    system("pause");
    return 0;
}
```

Output

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